Topic: Newton's Law of Cooling

Question: You prepare a cup of coffee and measure its temperature at $100°C$. After 2 minutes on the counter in your $25°C$ kitchen, the coffee has cooled to $90°C$. If you want to drink the coffee as soon as it cools to $50°C$, how long will you need to wait?

Answer choices:

A 10.30 minutes

B 15.35 minutes

C 12.20 minutes

D 25 minutes

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Solution: B

We'll list out what we know.

$T_0 = 100°$ Initial temperature of the coffee

$T_a = 25°$ Ambient temperature of the kitchen

$T(2) = 90°$ At time $t = 2$ minutes, the coffee has cooled to $90°$

If we plug everything we know into the Newton's Law of Cooling solution equation, we get

$$T(t) = T_a + (T_0 - T_a)e^{-kt}$$

$$T(t) = 25 + (100 - 25)e^{-kt}$$

$$T(t) = 25 + 75e^{-kt}$$

Substitute the initial condition $T(2) = 90°$,

$$T(2) = 25 + 75e^{-k(2)}$$

$$90 = 25 + 75e^{-2k}$$

in order to find a value for the decay constant k.

$$65 = 75e^{-2k}$$

$$\frac{13}{15} = e^{-2k}$$

$$\ln \frac{13}{15} = \ln(e^{-2k})$$

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$$\ln \frac{13}{15} = -2k$$

$$k = -\frac{1}{2} \ln \frac{13}{15}$$

Substitute this value for k into the equation modeling temperature over time.

$$T(t) = 25 + 75e^{-\left(-\frac{1}{2} \ln \frac{13}{15}\right)t}$$

$$T(t) = 25 + 75e^{\left(\frac{1}{2} \ln \frac{13}{15}\right)t}$$

We want to find the time t at which the soup reaches $50°$, so we'll substitute $T(t) = 50°$.

$$50 = 25 + 75e^{\left(\frac{1}{2} \ln \frac{13}{15}\right)t}$$

$$25 = 75e^{\left(\frac{1}{2} \ln \frac{13}{15}\right)t}$$

$$\frac{1}{3} = e^{\left(\frac{1}{2} \ln \frac{13}{15}\right)t}$$

Apply the natural logarithm to both sides of the equation.

$$\ln \frac{1}{3} = \ln \left(e^{\left(\frac{1}{2} \ln \frac{13}{15}\right)t} \right)$$

$$\ln \frac{1}{3} = \left(\frac{1}{2} \ln \frac{13}{15} \right) t$$

$$2 \ln \frac{1}{3} = \left(\ln \frac{13}{15} \right) t$$

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$$t = \frac{2 \ln \frac{1}{3}}{\ln \frac{13}{15}}$$

$t \approx 15.35$ minutes

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Topic: Newton's Law of Cooling

Question: You light a scented candle in a glass container. When you blow out the flame, the temperature of the glass is $200°C$. Room temperature is $20°C$ and after 5 minutes the glass has cooled to $150°C$. If it's safe to handle the glass at $40°C$, how long should you wait before touching the glass?

Answer choices:

A 16.88 minutes

B 22.79 minutes

C 50.24 minutes

D 33.76 minutes

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Solution: D

We'll list out what we know.

$T_0 = 200°$ Initial temperature of the glass

$T_a = 20°$ Ambient temperature in the room

$T(5) = 150°$ At time $t = 5$ minutes, the glass has cooled to $150°$

If we plug everything we know into the Newton's Law of Cooling solution equation, we get

$$T(t) = T_a + (T_0 - T_a)e^{-kt}$$

$$T(t) = 20 + (200 - 20)e^{-kt}$$

$$T(t) = 20 + 180e^{-kt}$$

Substitute the initial condition $T(5) = 150°$,

$$T(5) = 20 + 180e^{-k(5)}$$

$$150 = 20 + 180e^{-5k}$$

in order to find a value for the decay constant k.

$$130 = 180e^{-5k}$$

$$\frac{13}{18} = e^{-5k}$$

$$\ln \frac{13}{18} = \ln(e^{-5k})$$

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$$\ln \frac{13}{18} = -5k$$

$$k = -\frac{1}{5} \ln \frac{13}{18}$$

Substitute this value for k into the equation modeling temperature over time.

$$T(t) = 20 + 180e^{-\left(-\frac{1}{5} \ln \frac{13}{18}\right)t}$$

$$T(t) = 20 + 180e^{\left(\frac{1}{5} \ln \frac{13}{18}\right)t}$$

We want to find the time t at which the glass reaches $40°$, so we'll substitute $T(t) = 40°$.

$$40 = 20 + 180e^{\left(\frac{1}{5} \ln \frac{13}{18}\right)t}$$

$$20 = 180e^{\left(\frac{1}{5} \ln \frac{13}{18}\right)t}$$

$$\frac{1}{9} = e^{\left(\frac{1}{5} \ln \frac{13}{18}\right)t}$$

Apply the natural logarithm to both sides of the equation.

$$\ln \frac{1}{9} = \ln \left(e^{\left(\frac{1}{5} \ln \frac{13}{18}\right)t} \right)$$

$$\ln \frac{1}{9} = \left(\frac{1}{5} \ln \frac{13}{18} \right) t$$

$$5 \ln \frac{1}{9} = \left(\ln \frac{13}{18} \right) t$$

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$$t = \frac{5 \ln \frac{1}{9}}{\ln \frac{13}{18}}$$

$t \approx 33.76$ minutes

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Topic: Newton's Law of Cooling

Question: At a local tea shop, tea that's boiling at $120°C$ has just been removed from the heat and set out on the countertop, where the ambient temperature is $20°C$. After 3 minutes, the tea has cooled to $114°C$. If it's safe to drink the tea at $60°C$, how long do we need to wait to drink the tea?

Answer choices:

A 40.04 minutes

B 42.44 minutes

C 44.42 minutes

D 48.24 minutes

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Solution: C

We'll list out what we know.

$T_0 = 120°C$ Initial temperature of the tea

$T_a = 20°C$ Ambient temperature on the countertop

$T(3) = 114°C$ At time $t = 3$ minutes, the tea has cooled to $114°C$

If we plug everything we know into the Newton's Law of Cooling solution equation, we get

$$T(t) = T_a + (T_0 - T_a)e^{-kt}$$

$$T(t) = 20 + (120 - 20)e^{-kt}$$

$$T(t) = 20 + 100e^{-kt}$$

Substitute the initial condition $T(3) = 114°C$,

$$114 = 20 + 100e^{-3k}$$

in order to find a value for the decay constant k.

$$94 = 100e^{-3k}$$

$$0.94 = e^{-3k}$$

$$\ln 0.94 = \ln(e^{-3k})$$

$$\ln 0.94 = -3k$$

$$k = -\frac{1}{3}\ln 0.94$$

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Substitute this value for k into the equation modeling temperature over time.

$$T(t) = 20 + 100e^{-\left(-\frac{1}{3}\ln 0.94\right)t}$$

$$T(t) = 20 + 100e^{\frac{1}{3}t\ln 0.94}$$

We want to find the time t at which the temperature of the tea reaches $60°C$, so we'll substitute $T(t) = 60°$.

$$60 = 20 + 100e^{\frac{1}{3}t\ln 0.94}$$

$$40 = 100e^{\frac{1}{3}t\ln 0.94}$$

$$\frac{2}{5} = e^{\frac{1}{3}t\ln 0.94}$$

Apply the natural logarithm to both sides of the equation.

$$\ln\frac{2}{5} = \ln e^{\frac{1}{3}t\ln 0.94}$$

$$\ln\frac{2}{5} = \frac{1}{3}t\ln 0.94$$

$$t = \frac{3\ln\frac{2}{5}}{\ln 0.94}$$

$t \approx 44.42$ minutes

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